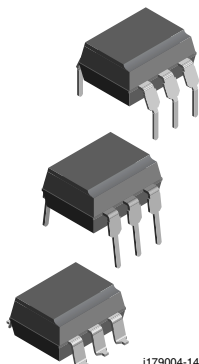
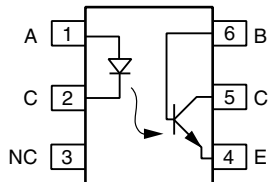




Optocoupler, Phototransistor Output, with Base Connection



i179004-14



FEATURES

- Isolation test voltage 5000 V_{RMS}
- Interfaces with common logic families
- Input-output coupling capacitance < 0.5 pF
- Industry standard dual-in-line 6-pin package
- Compliant to RoHS Directive 2002/95/EC and in accordance to WEEE 2002/96/EC

RoHS
COMPLIANT

APPLICATIONS

- AC mains detection
- Reed relay driving
- Switch mode power supply feedback
- Telephone ring detection
- Logic ground isolation
- Logic coupling with high frequency noise rejection

AGENCY APPROVALS

- UL file no. E52744
- cUL tested to CSA 22.2 bulletin 5A
- DIN EN 60747-5-2 (VDE 0884)/DIN EN 60747-5-5 (pending), available with option 1
- BSI: EN 60065, EN 60950-1
- FIMKO
- CQC

DESCRIPTION

The 4N25 family is an Industry Standard Single Channel Phototransistor Coupler. This family includes the 4N25, 4N26, 4N27, 4N28. Each optocoupler consists of gallium arsenide infrared LED and a silicon NPN phototransistor.

These couplers are Underwriters Laboratories (UL) listed to comply with a 5300 V_{RMS} isolation test voltage. This isolation performance is accomplished through special Vishay manufacturing process.

Compliance to DIN EN 60747-5-2 (VDE 0884)/DIN EN 60747-5-5 pending partial discharge isolation specification is available by ordering option 1.

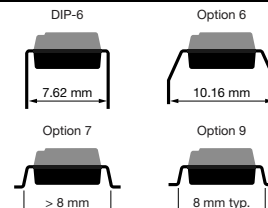
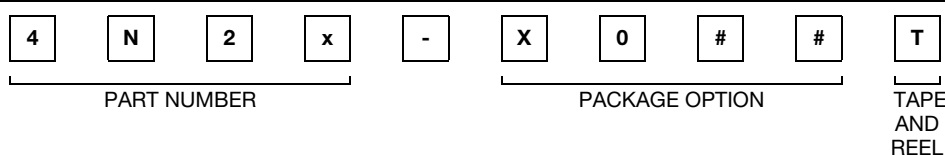
These isolation processes and the Vishay ISO9001 quality program results in the highest isolation performance available for a commercial plastic phototransistor optocoupler.

The devices are also available in lead formed configuration suitable for surface mounting and are available either on tape and reel, or in standard tube shipping containers.

Note

- For additional design information see application note 45 normalized curves

ORDERING INFORMATION



AGENCY CERTIFIED/PACKAGE	CTR (%)			
UL, cUL, BSI, FIMKO	≥ 20		≥ 10	
DIP-6	4N25-X000	-	4N27-X000	-
DIP-6, 400 mil, option 6	4N25-X006	4N26-X006	-	-
SMD-6, option 7	4N25-X007T	-	4N27-X007	-
SMD-6, option 9	4N25-X009T ⁽¹⁾	4N26-X009T ⁽¹⁾	4N27-X009T ⁽¹⁾	4N28-X009T ⁽¹⁾
VDE, UL, cUL, BSI, FIMKO	≥ 20		≥ 10	
DIP-6	4N25-X001	4N26-X001	-	4N28-X001
DIP-6, 400 mil, option 6	4N25-X016	4N26-X016	-	-
SMD-6, option 7	4N25-X017T ⁽¹⁾	4N26-X017T ⁽¹⁾	4N27-X017T	-

Notes

- Additional options may be possible, please contact sales office.
- ⁽¹⁾ Also available in tubes; do not put T on end.



ABSOLUTE MAXIMUM RATINGS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
INPUT				
Reverse voltage		V_R	6	V
Forward current		I_F	60	mA
Surge current	$t \leq 10\text{ }\mu\text{s}$	I_{FSM}	2.5	A
Power dissipation		P_{diss}	70	mW
OUTPUT				
Collector emitter breakdown voltage		V_{CEO}	70	V
Emitter base breakdown voltage		V_{EBO}	7	V
Collector current		I_C	50	mA
Collector peak current	$t_p/T = 0.5, t_p \leq 10\text{ ms}$	I_{CM}	100	mA
Output power dissipation		P_{diss}	150	mW
COUPLER				
Isolation test voltage		V_{ISO}	5000	V_{RMS}
Creepage distance			≥ 7	mm
Clearance distance			≥ 7	mm
Isolation thickness between emitter and detector			≥ 0.4	mm
Comparative tracking index	DIN IEC 112/VDE0303, part 1		≥ 175	
Isolation resistance	$V_{IO} = 500\text{ V}, T_{amb} = 25\text{ }^{\circ}\text{C}$	R_{IO}	$\geq 10^{12}$	Ω
	$V_{IO} = 500\text{ V}, T_{amb} = 100\text{ }^{\circ}\text{C}$	R_{IO}	$\geq 10^{11}$	Ω
Storage temperature		T_{stg}	- 55 to + 150	$^{\circ}\text{C}$
Operating temperature		T_{amb}	- 55 to + 100	$^{\circ}\text{C}$
Junction temperature		T_j	100	$^{\circ}\text{C}$
Soldering temperature ⁽¹⁾	2 mm from case, $\leq 10\text{ s}$	T_{sld}	260	$^{\circ}\text{C}$

Notes

- Stresses in excess of the absolute maximum ratings can cause permanent damage to the device. Functional operation of the device is not implied at these or any other conditions in excess of those given in the operational sections of this document. Exposure to absolute maximum ratings for extended periods of the time can adversely affect reliability.
- Refer to reflow profile for soldering conditions for surface mounted devices (SMD). Refer to wave profile for soldering conditions for through hole devices (DIP).

ELECTRICAL CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)							
PARAMETER	TEST CONDITION	PART	SYMBOL	MIN.	TYP.	MAX.	UNIT
INPUT							
Forward voltage ⁽¹⁾	$I_F = 50\text{ mA}$		V_F		1.36	1.5	V
Reverse current ⁽¹⁾	$V_R = 3.0\text{ V}$		I_R		0.1	100	μA
Capacitance	$V_R = 0\text{ V}$		C_O		25		pF
OUTPUT							
Collector base breakdown voltage ⁽¹⁾	$I_C = 100\text{ }\mu\text{A}$		BV_{CBO}	70			V
Collector emitter breakdown voltage ⁽¹⁾	$I_C = 1.0\text{ mA}$		BV_{CEO}	30			V
Emitter collector breakdown voltage ⁽¹⁾	$I_E = 100\text{ }\mu\text{A}$		BV_{ECO}	7			V
$I_{CEO}(\text{dark})$ ⁽¹⁾	$V_{CE} = 10\text{ V}, (\text{base open})$	4N25			5	50	nA
		4N26			5	50	nA
		4N27			5	50	nA
		4N28			10	100	nA
$I_{CBO}(\text{dark})$ ⁽¹⁾	$V_{CB} = 10\text{ V}, (\text{emitter open})$				2.0	20	nA
Collector emitter capacitance	$V_{CE} = 0$		C_{CE}		6.0		pF
COUPLER							
Isolation test voltage ⁽¹⁾	Peak, 60 Hz		V_{IO}	5000			V
Saturation voltage, collector emitter	$I_{CE} = 2.0\text{ mA}, I_F = 50\text{ mA}$		$V_{CE(\text{sat})}$			0.5	V
Resistance, input output ⁽¹⁾	$V_{IO} = 500\text{ V}$		R_{IO}	100			$G\Omega$
Capacitance, input output	$f = 1\text{ MHz}$		C_{IO}		0.5		pF

Notes

- Minimum and maximum values are testing requirements. Typical values are characteristics of the device and are the result of engineering evaluation. Typical values are for information only and are not part of the testing requirements.
- JEDEC registered values are 2500 V, 1500 V, 1500 V and 500 V for the 4N25, 4N26, 4N27, and 4N28 respectively.

CURRENT TRANSFER RATIO ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

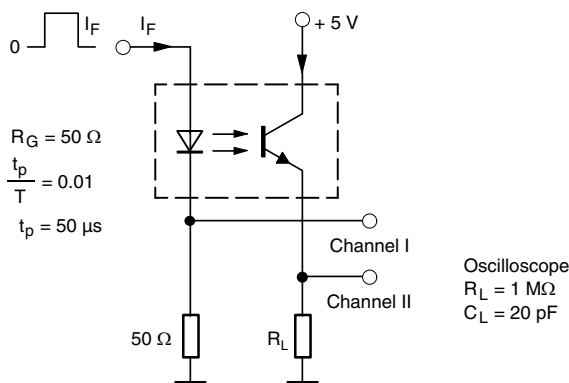
PARAMETER	TEST CONDITION	PART	SYMBOL	MIN.	TYP.	MAX.	UNIT
I_C/I_F	$V_{CE} = 10\text{ V}$, $I_F = 10\text{ mA}$	4N25	CTR_{DC}	20	50		%
		4N26	CTR_{DC}	20	50		%
		4N27	CTR_{DC}	10	30		%
		4N28	CTR_{DC}	10	30		%

Note

- Indicates JEDEC registered values.

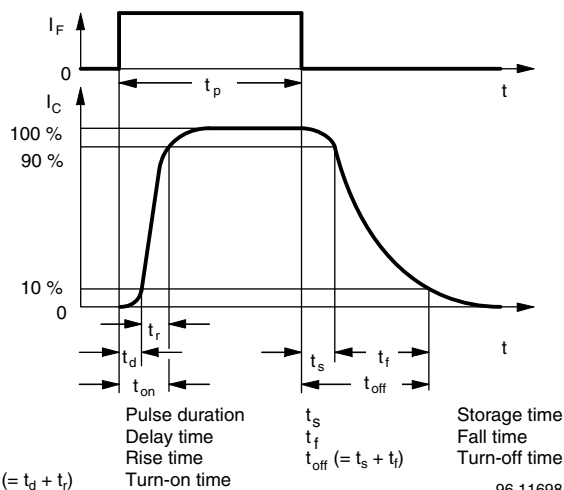
SWITCHING CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	MIN.	TYP.	MAX.	UNIT
Rise time	$V_{CC} = 5\text{ V}$, $I_F = 10\text{ mA}$, $R_L = 100\text{ }\Omega$	t_r		2.0		μs
Fall time	$V_{CC} = 5\text{ V}$, $I_F = 10\text{ mA}$, $R_L = 100\text{ }\Omega$	t_f		2.0		μs



95 10804-3

Fig. 1 - Test Circuit, Non-Saturated Operation



96 11698

Fig. 2 - Switching Times

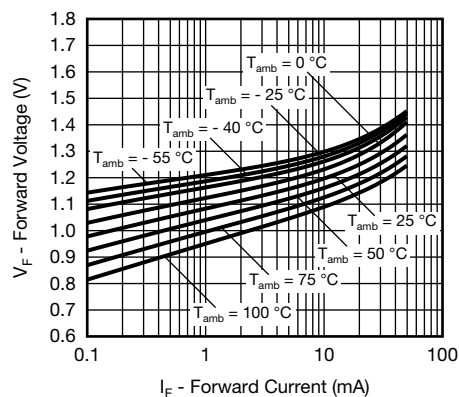
TYPICAL CHARACTERISTICS ($T_{amb} = 25\text{ }^{\circ}\text{C}$, unless otherwise specified)


Fig. 3 - Forward Voltage vs. Forward Current

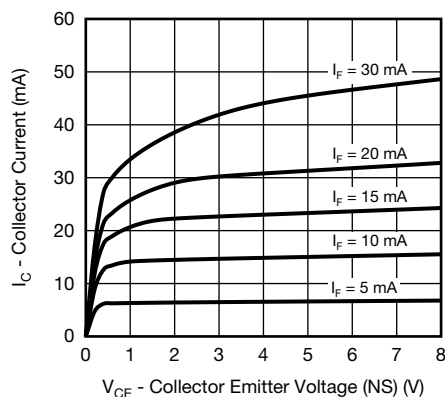


Fig. 4 - Collector Current vs. Collector Emitter Voltage (NS)

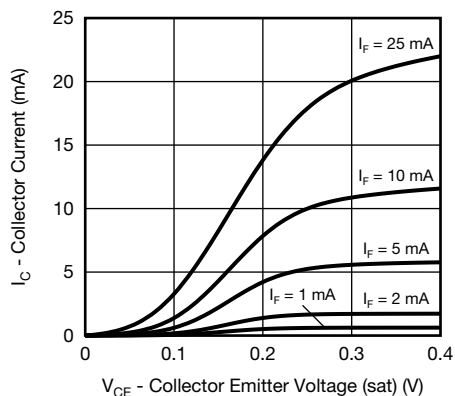


Fig. 5 - Collector Current vs. Collector Emitter Voltage (sat)

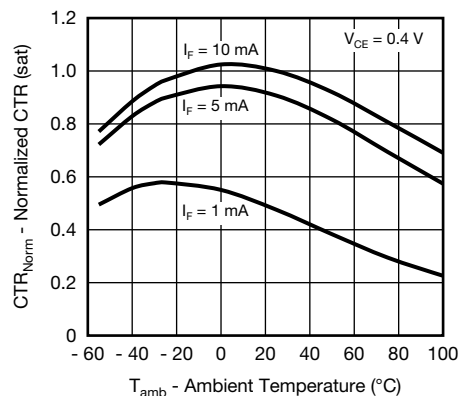


Fig. 8 - Normalized CTR (sat) vs. Ambient Temperature

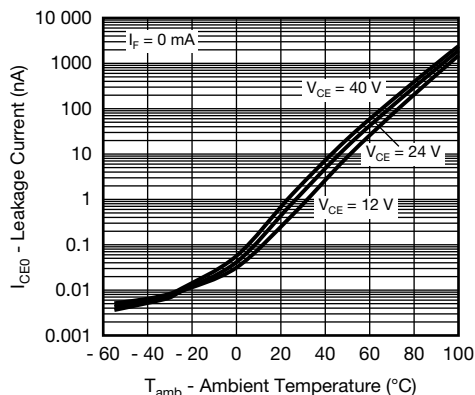


Fig. 6 - Leakage Current vs. Ambient Temperature

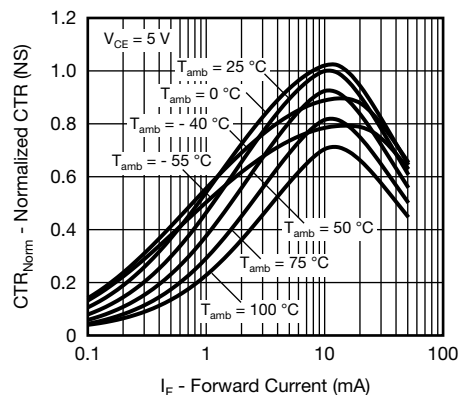


Fig. 9 - Normalized CTR (NS) vs. Forward Current

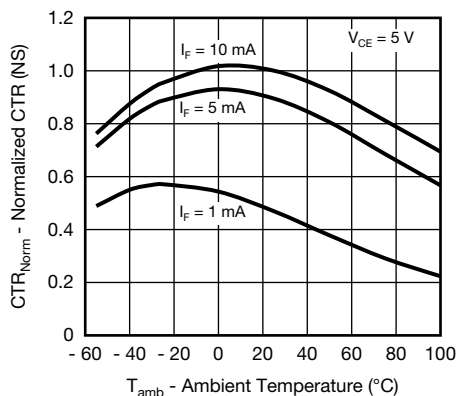


Fig. 7 - Normalized CTR (NS) vs. Ambient Temperature

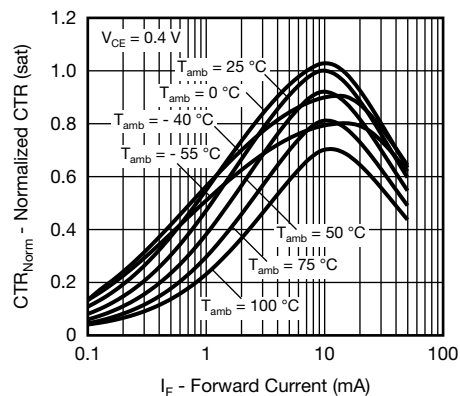


Fig. 10 - Normalized CTR (sat) vs. Forward Current

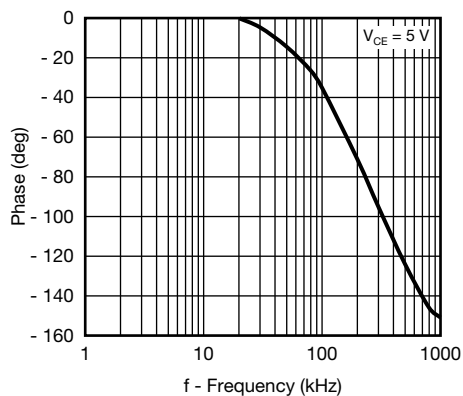


Fig. 11 - CTR Frequency vs. Phase Angle

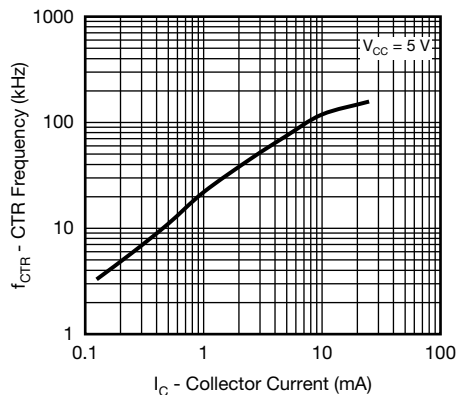


Fig. 12 - CTR Frequency vs. Collector Current

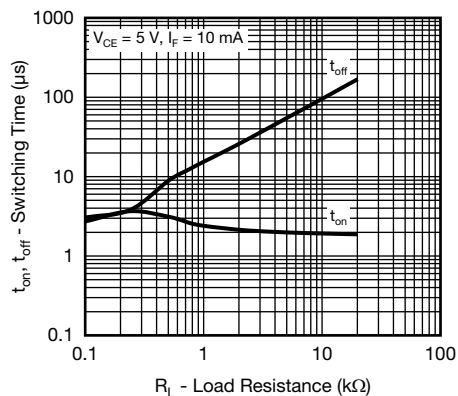
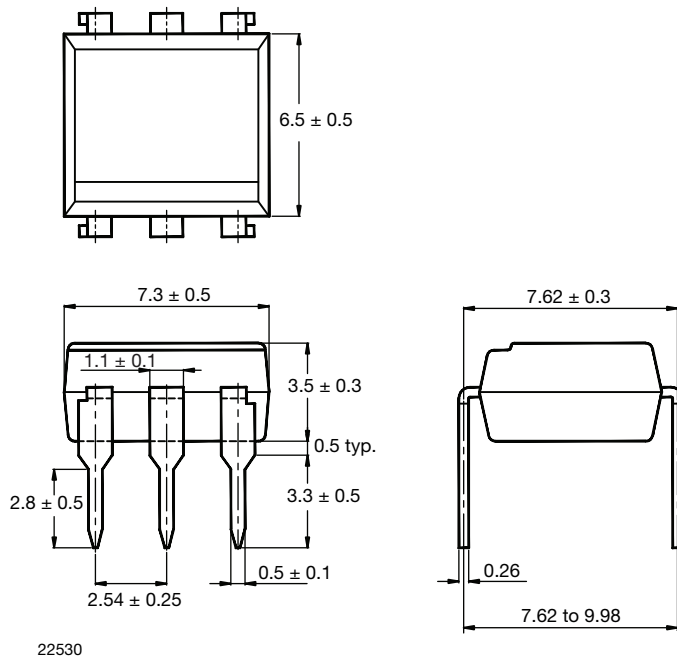
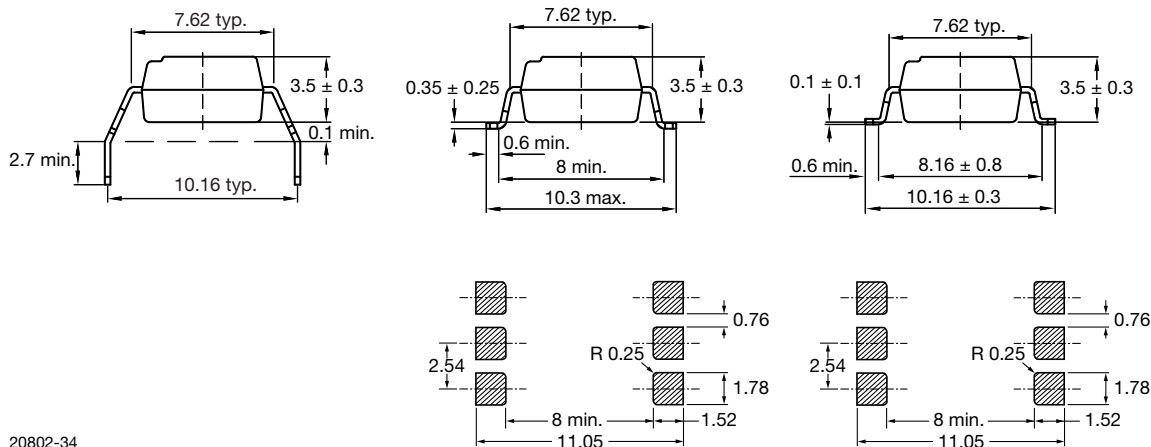
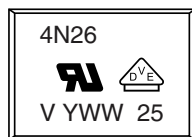


Fig. 13 - Switching Time vs. Load Resistance

PACKAGE DIMENSIONS in millimeters

Option 6
Option 7
Option 9

PACKAGE MARKING

Notes

- VDE logo is only marked on option 1 parts. Option information is not marked on the part.
- Tape and reel suffix (T) is not part of the package marking.



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